

PINNFORGE: EXECUTION-GROUNDED EVOLUTION- ARY DESIGN OF PHYSICS-INFORMED NEURAL NET- WORKS FOR PDE SOLVING VIA LARGE LANGUAGE MODELS SUPPLEMENTARY MATERIAL

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A METHOD IMPLEMENTATION AND SEARCH MECHANICS

This appendix specifies the executable candidate representation, LLM context, role-conditioned search directions, controller-side validation, ablation isolation, and a representative cross-generation trace. It also separates decisions made by the LLM from those fixed by the deterministic controller.

A.1 ALGORITHM SPEC AND EXECUTABLE DESIGN DIMENSIONS

PINNForge represents each candidate as a structured `AlgorithmSpec` rather than arbitrary executable code. The LLM proposes controller-supported components, and a candidate is executed only after deterministic normalization, compatibility checks, and budget validation. We denote the controller-supported configuration space by Ω .

The `AlgorithmSpec` jointly covers the main PINN design dimensions. The table below lists representative executable choices exposed by the implementation; these entries describe each dimension’s scope rather than a fixed PDE-specific template.

Design dimension	Representative executable choices
Representation	Coordinate transforms, Fourier or periodic features, and multiscale representations supported by the implementation.
Architecture	Network depth and width, activation functions, residual connections, and other supported topology-related choices.
Sampling	Collocation-point allocation, sampling distributions, and adaptive or localized resampling mechanisms.
Constraints	Soft constraints, exact constraint transforms when available, and supported periodic or interface handling.
Loss	PDE-residual construction, loss weighting, normalization, causal weighting, and other supported physical-loss handling.
Optimization	Optimizer choice, learning rate, gradient handling, scheduling, and staged optimization.
Training control	Resampling, curriculum or stage allocation, checkpoint-related control, and other supported training procedures.

These dimensions are searched jointly. A proposal may preserve some components while modifying several coupled components at once—for example, representation, sampling, and optimization—when the observed training behavior suggests that a scalar adjustment is insufficient. Every candidate remains restricted to controller-supported implementations in Ω .

The controller also enforces resource limits that the LLM cannot relax: at most 2M trainable parameters, 65,536 training points, and fixed low- and high-fidelity budgets of 1K and 10K optimizer steps, respectively (Appendix B). Thus, Ω defines the admissible design space, while the experimental protocol defines its resource envelope.

A.2 LLM INPUTS, OUTPUTS, AND INFORMATION BOUNDARIES

Each LLM call proposes one complete structured `AlgorithmSpec`; it does not generate arbitrary executable code. A controller handles parsing, validation, instantiation, and real PINN training, preventing the LLM from bypassing the search space or modifying the experimental protocol.

The information provided to the LLM at different search stages is summarized below.

Context	Generation 0	Generation $g \geq 1$	Purpose
Target PDE description	✓	✓	Provides the governing equation, computational domain, IC/BC, and problem characteristics
Prior knowledge $K_{\mathcal{P}}$	✓	✓	Provides PINN design rationale relevant to the current PDE
Allowed search space Ω	✓	✓	Restricts the network, sampling, constraint, loss, and optimization components that may be generated
Training / resource budget	✓	✓	Restricts parameter count, sampling scale, and training budget
Role / search-strategy condition	✓	✓	Specifies the design focus of the current candidate
Global Top-3 <code>AlgorithmSpec</code>	–	✓	Provides validated high-performing parent configurations
Candidate-specific execution evidence of parents	–	✓	Provides raw parent-linked diagnostics including PDE residuals, IC/BC errors, optimization stability, and convergence behavior
Run-scoped posterior memory M_g	–	✓	Provides a compact cross-candidate summary of successes, failures, and design experience observed within the current independent search run

Generation 0 has no execution history, so initial candidates use only PDE information, prior knowledge, the allowed search space, resource constraints, and initialization roles. From Generation 1 onward, the context also includes the Global Top-3 parents, their execution evidence, and the run-scoped posterior memory. Appendix A.4 defines the roles and evolutionary strategies; Appendix B.2 gives their sampling temperatures.

The LLM cannot modify the target PDE, domain, initial or boundary conditions, evaluation protocol, or experimental budget. Its output is restricted to PINN components in Ω , including representation, architecture, activation, constraints, sampling, loss handling, optimization, and training control. Candidates are normalized before deduplication; exactly identical normalized `AlgorithmSpecs` are treated as one configuration, avoiding duplicates caused only by field ordering or expanded defaults.

A.3 PDE-RELATED PRIOR KNOWLEDGE AND RUN-SCOPED EXECUTION MEMORY

PINNForge separates information available before execution from empirical evidence collected during the current search run. This distinction keeps general design rationale separate from problem-specific training experience.

PDE-related prior knowledge. For a target PDE $\mathcal{P}$, PINNForge characterizes properties relevant to PINN design, including time dependence, periodicity, multiscale structure, geometric complexity, locally steep behavior, and physical-constraint types. Associated design knowledge is supplied as $K_{\mathcal{P}}$, which provides physical and numerical rationale for design hypotheses without prescribing a specific architecture, optimizer, sampling rule, or complete solver. The same prior remains available in later generations alongside execution evidence.

Candidate-level execution evidence. After a candidate is trained on $\mathcal{P}$, execution returns (y_i, E_i) . Here, y_i is the scalar comparison metric used for ranking, while E_i records diagnostics such as PDE residual, constraint satisfaction, loss reduction, stagnation, gradient or loss imbalance when available, convergence behavior, and numerical stability. Keeping y_i separate from E_i distinguishes ranking from the diagnostic context used to guide later design changes.

Run-scoped posterior memory. Let $\mathcal{H}_g$ be the complete history through Generation g , including validated `AlgorithmSpecs` and execution results. A separate posterior-summary LLM call compresses recurring successes, failures, physical-constraint behavior, and optimization experience in $\mathcal{H}_g$ into run-scoped memory M_g . This call only summarizes evidence from the current independent run; it neither generates a candidate nor changes controller-side rankings. Unlike $K_{\mathcal{P}}$, M_g contains no experience from other runs and is absent at Generation 0.

Candidate-specific evidence and posterior memory therefore serve different roles. For Generation $g + 1$, $E_{P_{g+1}}$ provides detailed evidence for the selected Global Top-3 parents, whereas M_g summarizes the broader history $\mathcal{H}_g$. After Generation $g + 1$ executes, the history becomes $\mathcal{H}_{g+1}$ and the posterior-summary call produces M_{g+1} . Thus, $K_{\mathcal{P}}$, parent evidence, and M_g provide prior, local, and run-level context at distinct granularities. Appendix B.4 specifies the controller-defined evaluation protocol.

A.4 INITIALIZATION ROLES AND EVOLUTION STRATEGIES

Generation-0 initialization roles. Generation 0 uses eight complementary roles to broaden initial coverage. Each role sets the design emphasis of one LLM call rather than a fixed solver template. All roles receive the same PDE description, $K_{\mathcal{P}}$, allowed design space, and resource constraints, and each produces one complete `AlgorithmSpec`.

Generation-0 role	Primary design emphasis
Robust reference	Construct a conservative reference design that prioritizes broadly applicable mechanisms and stable training.
Constraint / causality	Emphasize physical-constraint enforcement and, when relevant, temporal or causal structure.
Localized sampling	Emphasize local structure, collocation-point placement, and adaptive or localized refinement of difficult regions.
Representation	Emphasize network input or feature representation and its interaction with the solver architecture.
Residual / loss	Emphasize PDE-residual construction, physical-loss formulation, and loss-handling choices.
Optimization	Emphasize optimizer selection, scheduling, gradient behavior, and numerical training stability.
Capacity / topology	Emphasize model capacity, network depth and width, and supported topology-related choices.
Cross-component novelty	Explore less conventional combinations spanning multiple executable PINN components.

The eight candidates form the knowledge-guided initial population for subsequent execution-grounded evolution.

Evolution strategies for $g \geq 1$. After execution, PINNForge maintains the distinct Global Top-3 over the full search history. From Generation 1 onward, each new candidate is conditioned on these parents, their candidate-specific evidence, $K_{\mathcal{P}}$, M_g , and one of five evolutionary strategies. For strategy s ,

$$a_{g+1}^{(s)} = \text{LLMGenerate}(P_{g+1}, E_{P_{g+1}}, K_{\mathcal{P}}, M_g, s),$$

where P_{g+1} is the Global Top-3 parent set, $E_{P_{g+1}}$ is its candidate-specific execution evidence, and M_g summarizes the broader history $\mathcal{H}_g$.

Evolution strategy	Primary evolution direction
Refine	Use the current strongest candidate as the primary parent, preserve effective existing mechanisms, and make limited adjustments targeting its main shortcomings.
Physics-Guided	Adjust sampling-, constraint-, and loss-related designs primarily according to the observed PDE residual and initial/boundary-condition errors.
Architecture-Guided	Adjust network representation, architecture, and related training configurations according to representation capacity, model capacity, and optimization behavior.
Synthesis	Combine complementary, well-performing design mechanisms from multiple Global Top-3 parents into a new complete configuration.
Novelty	Explore design combinations not yet covered by the current high-performing candidates or the existing search trajectory.

The strategies define semantic search preferences rather than field-wise mutation or crossover operators. The LLM decides what to preserve, modify jointly, or recombine from multiple parents using PDE characteristics and observed execution behavior. After execution, new results extend the history to $\mathcal{H}_{g+1}$, the posterior-summary call updates M_g to M_{g+1} , and all candidates compete for the Global Top-3. This closes the generate–execute–evaluate–evolve loop.

The role- and strategy-specific sampling temperatures, including the more exploratory escape-generation schedule, are listed in Appendix B.2.

A.5 DETERMINISTIC VALIDATION AND NORMALIZATION

The LLM only proposes candidates; a deterministic controller decides whether each proposal is admitted to training. Every returned `AlgorithmSpec` passes through

Parsing $\rightarrow$ Normalization $\rightarrow$ Validation $\rightarrow$ Instantiation $\rightarrow$ Training.

The validation stage checks at least the following conditions:

- whether the structure of the `AlgorithmSpec` is complete;
- whether all components belong to the allowed search space Ω ;
- whether different algorithm components are mutually compatible;
- whether the number of network parameters satisfies the budget;
- whether the sampling budget satisfies the constraints;
- whether the training iteration count strictly matches the current low- or high-fidelity training budget.

Only candidates that pass deterministic validation are instantiated and trained under the experimental protocol, producing $(y_{g,k}, E_{g,k})$ for later search. LLM suggestions alone are not valid candidates: only `AlgorithmSpecs` that pass deterministic validation and complete real PINN training enter the history and Global Top-3.

A.6 INFORMATION ISOLATION IN ABLATION EXPERIMENTS

Section 4.2 defines three ablations. To focus each intervention on designated information channels, we keep the candidate count, training budget, search space, and LLM generation protocol unchanged while varying the context available to the LLM.

Table A.1: Information accessible to the LLM under different ablation settings.

Variant	Prior Knowledge	Run-scoped Posterior	Execution Feedback to LLM	Parent-based Evolution
w/o Knowledge Guidance	×	×	✓	✓
w/o Execution Feedback to LLM	✓	×	×	✓
w/o Evolutionary Search	✓	✓	✓	×
PINNForge	✓	✓	✓	✓

In w/o Knowledge Guidance, the LLM receives neither $K_{\mathcal{P}}$ nor M_g , while raw execution evidence and parent-based evolution are retained. This variant therefore evaluates the knowledge-guidance channel as a whole rather than isolating static prior knowledge from the execution-derived posterior.

In w/o Execution Feedback, historical candidate structures remain visible, but $y_{g,k}$, $E_{g,k}$, residuals, constraint errors, optimization diagnostics, rankings, and other training outcomes are hidden. Because M_g depends on this history, no posterior memory is formed. The LLM therefore retains prior knowledge and parent structures without access to their observed execution behavior.

In w/o Evolutionary Search, the LLM retains aggregated historical execution experience but receives no historical AlgorithmSpecs or Global Top-3 parents. Candidates are generated independently, without inheriting, modifying, or recombining existing high-performing configurations, under the same candidate-evaluation budget.

A.7 EXECUTION FEEDBACK AND CROSS-GENERATION INHERITANCE EVIDENCE ON BURGERS-1D

The main text gives a compact Burgers-1D parent–child example and shows that later generations continue to find lower-MSE candidates after Generation 0; Appendix C reports all 25 trajectories. Here we provide the full component-level trace and the final candidate’s cross-generation lineage, showing how training feedback informs later design changes and how parent inheritance contributes to the final configuration.

Below, we use Gg–Ck as a readable label for candidate $a_{g,k}$; for example, G6–C4 corresponds to $a_{6,4}$.

A.7.1 Representative Parent—Child Modification For parent G6–C4, the observed run shows numerically stable optimization and very small hard-constraint errors, but a clear PDE-residual hotspot remains near the shock. The child preserves the stable network and constraint mechanisms, strengthens local adaptive sampling, and explores an alternative optimization configuration.

Table A.2: Representative execution-feedback-driven modification on Burgers-1D.

Component	Parent Configuration	Execution Evidence	Design Decision	Child Configuration
Architecture	Modified MLP, 6×128 ; no residual connections	Completed the full 1,000-step training without NaN or solution collapse	Preserve. The architecture was numerically stable in the observed run	Keep Modified MLP, 6×128
Constraint	Exact hard transform burgers_1d_initBCerror	IC error = 2.25×10^{-8} , BC error = 8.74×10^{-8}	Preserve. Do not reallocate search emphasis to constraints that are already well satisfied	Keep the exact hard transform

Component	Parent Configuration	Execution Evidence	Design Decision	Child Configuration
Sampling	10,000 Sobol interior points + RAD; add 500 each time; cap at 15,000; residual exponent 2.5	Persistent residual hotspot at $x \in [-0.24, 0.24]$, $t \in [0.15, 0.39]$	Modify. Strengthen local refinement while preserving global Sobol coverage	Add 600 each time; raise the cap to 16,000; keep the remaining RAD settings unchanged
Optimization	Adam 800 steps, lr 1.0×10^{-3} , one-cycle; L-BFGS 200 steps, history 100	Optimization remains numerically stable, but clear residual error remains at the end of training	Explore. Treat optimization changes as accompanying exploration rather than misclassifying a stable optimizer as a failure	Increase Adam lr to 1.2×10^{-3} , switch to step decay; L-BFGS history 120

The actual training results of the parent and child are:

Metric	Parent G6-C4	Child G7-C4
Solution MSE	5.71×10^{-4}	2.21×10^{-4}
PDE residual	6.05×10^{-2}	2.14×10^{-2}
IC error	2.25×10^{-8}	1.25×10^{-8}
BC error	8.74×10^{-8}	2.53×10^{-8}

This example does not attribute the improvement to a single component: parent and child use independent random initializations, and both sampling and optimization change. Instead, it records how execution feedback distinguishes mechanisms to preserve from components that remain candidates for modification.

A.7.2 Cross-Generation Evolutionary Lineage of the Final Candidate PINNForge records the parent source of each evolutionary candidate. Single-parent refinement is written as

$$a_{g,p} \rightarrow a_{g+1,c},$$

Multi-parent synthesis inherits and recombines mechanisms from several Global Top-3 candidates. Refinement and synthesis are not predefined field-wise operators; the LLM decides what to preserve, modify, or combine from the parent `AlgorithmSpecs`, target PDE, and execution evidence.

Table A.3: Cross-generation evolutionary lineage of the final Burgers-1D candidate.

Generation	Candidate	Parent(s)	Evolution Mode	Main Inherited / Modified Components
0	G0-C2	–	Independent Initialization	4×64 modified MLP, $\tanh$, soft constraints, Sobol + RAD, Adam + L-BFGS

Generation	Candidate	Parent(s)	Evolution Mode	Main Inherited / Modified Components
1	G1-C3	G0-C2, G0-C6, G0-C4	Multi-Parent Synthesis	Combine RAD sampling, a deeper network topology, and optimization control; expand to a 6×96 modified MLP
2	G2-C1	G1-C3	Single-Parent Refinement	Preserve the 6×96 network; increase the RAD exponent and interior-point budget
3	G3-C4	G2-C1, G2-C4	Multi-Parent Synthesis	Introduce an exact hard IC/BC transform; use fixed weighting; increase the RAD cap
4	G4-C4	G3-C4, G3-C2	Multi-Parent Synthesis	Preserve hard constraints and Sobol + RAD; widen the network to 6×128 ; add one-cycle Adam and L-BFGS recovery
5	G5-C4	G4-C4, G4-C3	Multi-Parent Synthesis	Adjust the Adam schedule, RAD capacity, and L-BFGS history
6	G6-C4	G5-C4, G5-C1, G5-C3	Multi-Parent Synthesis	Preserve hard constraints, $\tanh$, Sobol + RAD, and Adam $\rightarrow$ L-BFGS; use a stable 6×128 modified MLP
7	G7-C4	G6-C4	Single-Parent Refinement	Strengthen RAD refinement; adjust the Adam schedule and L-BFGS history

Generation	Candidate	Parent(s)	Evolution Mode	Main Inherited / Modified Components
8	G8-C4	G7-C1, G7-C2, G7-C4	Multi-Parent Synthesis	Combine network capacity, sampling, and training strategies from multiple G7 parents; use a 5×192 modified MLP and retune RAD and L-BFGS settings
Final HF	G8-C4	Low-fidelity source: G8-C4	Independent High-Fidelity Training	Keep the G8-C4 AlgorithmSpec unchanged and independently train for 10,000 optimizer steps from a new random initialization

The lineage shows that the final configuration is assembled across generations by inheriting, modifying, and synthesizing mechanisms from high-performing candidates rather than by a single independent LLM call. Together with the search trajectories, it provides a traceable chain from **execution evidence** $\rightarrow$ **targeted modification** $\rightarrow$ **parent inheritance** $\rightarrow$ **final configuration**.

B EXPERIMENTAL PROTOCOL AND COMPLETE RESULTS

This appendix provides the experimental protocol and complete results. Notation for candidates, execution evidence, search history, and the Global Top-3 follows Section 3 and Appendix A.

B.1 SEARCH CONFIGURATION

PINNForge starts with eight Generation-0 candidates and adds five candidates per later generation. Indices begin at $g = 0$; search completes at least Generations 0–3 and at most Generations 0–9, i.e., four to ten completed generations. After each generation, the controller updates the distinct all-history Global Top-3 defined in Section 3.3.

Setting	Value
Generation-0 candidates	8
Candidates per later generation	5
Min. / max. completed generations	4 / 10
Parent population	Global Top-3
Low-fidelity training	1K steps
High-fidelity training	10K steps
Maximum trainable parameters	2M
Maximum training points	65,536

All normal and escape generations count toward the ten-generation maximum.

B.2 LLM SAMPLING

PINNForge uses Qwen3.8-27B as the fixed LLM backbone for candidate generation, execution-feedback analysis, and evolutionary search, without PDE-specific fine-tuning. Candidate generation follows a role-conditioned temperature schedule: lower temperatures favor conservative directions, while higher temperatures encourage exploration.

Generation-0 direction	Temperature	
Robust reference	0.20	
Constraint / causality	0.35	
Localized sampling	0.40	
Representation	0.45	
Residual / loss	0.45	
Optimization	0.40	
Capacity / topology	0.50	
Cross-component novelty	0.60	

Evolution strategy	Normal	Escape
Refine	0.20	0.35
Physics-Guided	0.36	0.51
Architecture-Guided	0.42	0.57
Synthesis	0.47	0.62
Novelty	0.58	0.73

An escape generation raises each strategy temperature by 0.15, capped at 0.85. The posterior-summary call that constructs M_g uses temperature 0.1. `top_p` follows the provider default, and no provider-side sampling seed is supplied. Repeated executions therefore need not reproduce the same candidate sequence; the ten-run results summarize end-to-end stochasticity rather than one deterministic LLM trajectory.

B.3 COMPUTATIONAL ENVIRONMENT

Experiments are conducted on a Linux server using NVIDIA GeForce RTX 4090 GPUs. Each PDE search runs on a single GPU using FP32 arithmetic.

Component	Configuration
GPU	NVIDIA GeForce RTX 4090
NVIDIA Driver	550.163.01
CUDA	12.4
Python	3.12
PyTorch	2.6.0+cu124
Precision	FP32

Additional software dependencies and exact environment specifications are provided with the implementation.

B.4 EVALUATION AND REPORTING PROTOCOL

For each PDE, solution accuracy is measured by MSE on fixed reference evaluation points. No separate held-out reference subset is used. PINNForge uses the same points for low-fidelity comparison, Global Top-3 updates, stagnation and escape decisions, high-fidelity selection, and reporting.

Main results report MSE on these points over ten independent runs. Each run is one complete execution of the corresponding method, including its internal generation and selection procedure. RoPINN, PINNsFormer, and PINNsAgent use the same reference-point protocol.

Each valid low-fidelity candidate is trained for 1,000 optimizer steps. The final Global Top-3 configurations are then kept fixed, reinitialized, and independently trained from scratch for 10,000 steps; no low-fidelity parameters are reused. The configuration with the lowest high-fidelity evaluation MSE is selected.

B.5 DETERMINISTIC STAGNATION DETECTION, ESCAPE GENERATION, AND TERMINATION

All search-control decisions in this section are computed from the low-fidelity candidate comparison metric defined in Section 3.2 of the main paper. For candidate $a_{g,k}$, let $y_{g,k}$ denote its low-fidelity comparison value. In the reported experiments, $y_{g,k}$ is the solution MSE computed on the fixed reference evaluation points defined in Appendix B.4. Lower values are better. For each completed generation (g), define the best valid candidate value in that generation as

$$b_g = \min_k y_{g,k},$$

where the minimum is taken over valid candidates in Generation g . Define the all-history best value through Generation g as

$$B_g = \min_{0 \leq t \leq g} b_t.$$

When Generation g contains at least three valid candidates, define the median of its three best candidate values as

$$m_g = \text{Median}(\text{Top3}\{y_{g,k}\}_k).$$

The median comparison value of the distinct all-history Global Top-3 archive after completing Generation g is defined as

$$A_g = \text{Median}(\{y(a) : a \in \text{GlobalDistinctTop3}(\mathcal{H}_g)\}).$$

This quantity is used once at least three distinct valid candidates exist in the archive; before that point, archive-median criteria are disabled.

The relative improvement from an earlier value x to a later value y is defined as

$$I(x, y) = \begin{cases} \max\left(0, \frac{x - y}{|x|}\right), & x \neq 0 \text{ and both } x, y \text{ are finite,} \\ 0, & \text{otherwise.} \end{cases}$$

With this definition, larger $I(x, y)$ indicates a larger reduction in the comparison metric.

Stagnation detection begins after Generation 3 has completed, i.e., after four completed generations (Generations 0–3). For a normal generation $g \geq 3$, an escape generation is scheduled when at least one of the following conditions holds.

1. **Global-best stagnation.** The three most recent generation-to-generation improvements of the all-history best are all below 1%, and the cumulative improvement over the same window is below 2%:

$$I(B_{g-3}, B_{g-2}) < 0.01, \quad I(B_{g-2}, B_{g-1}) < 0.01,$$

$$I(B_{g-1}, B_g) < 0.01, \quad I(B_{g-3}, B_g) < 0.02.$$

2. **Elite-archive stagnation.** Across the last three completed generations, no previously unseen normalized `AlgorithmSpec` enters the distinct all-history Global Top-3 archive.

3. **Population stagnation.** When each generation in the relevant window contains at least three valid candidates, the median of the three best candidates improves by less than 2% in two consecutive transitions:

$$I(m_{g-2}, m_{g-1}) < 0.02, \quad I(m_{g-1}, m_g) < 0.02.$$

If stagnation is detected after Generation g_{stall} , and the 10-generation limit has not yet been reached, the next generation is designated as the escape generation:

$$g_{\text{esc}} = g_{\text{stall}} + 1.$$

The escape generation uses the same five semantic evolution strategies and the same 1,000-step low-fidelity training budget as a normal evolution generation, but increases exploration through the following deterministic controls:

- each evolution-strategy temperature is increased by 0.15, capped at 0.85;
- every generated candidate must differ from its parent context in at least two major executable modules;
- exact reproductions of parent configurations and non-substantive numerical perturbations are rejected;
- previously failed mechanisms are not repeated unless the candidate explicitly addresses the observed failure reason;
- Synthesis and Novelty are prompted to use greater parent or mechanism diversity.

The normal evolution-strategy temperature vector is

$$\tau_{\text{normal}} = (0.20, 0.36, 0.42, 0.47, 0.58),$$

and the corresponding escape-generation vector is

$$\tau_{\text{escape}} = (0.35, 0.51, 0.57, 0.62, 0.73).$$

After completing the escape generation g_{esc} , the escape is considered successful if at least one of the following conditions holds.

1. **New best candidate.** The best candidate in the escape generation improves on the pre-escape all-history best by at least 2%:

$$I(B_{g_{\text{stall}}}, b_{g_{\text{esc}}}) \geq 0.02.$$

2. **New elite configuration.** At least one previously unseen normalized `AlgorithmSpec` from Generation g_{esc} enters the distinct all-history Global Top-3 archive.
3. **Elite-archive improvement.** The median comparison value of the distinct Global Top-3 improves by at least 2%:

$$I(A_{g_{\text{stall}}}, A_{g_{\text{esc}}}) \geq 0.02.$$

If the escape succeeds, the stagnation history accumulated before g_{esc} is cleared and normal evolution resumes. If none of the three escape-success conditions is satisfied, the low-fidelity search terminates immediately and the current Global Top-3 is passed to the high-fidelity evaluation stage.

The low-fidelity search also terminates when the search has completed the maximum of 10 generations, i.e., after Generation 9 has finished.

Normal and escape generations both count toward the 10-generation maximum. Because generation indices run from 0 to 9 and stagnation is first assessed after Generation 3, the earliest possible escape generation is Generation 4.

B.6 COMPLETE MAIN RESULTS

Table 13 reports mean $\pm$ standard-deviation MSE over ten runs. Rank is PINNForge’s per-task rank among the four unified methods. PINNacle Best is included only as an oracle-style reference and is excluded from the Friedman and pairwise significance tests.

Table 13: Complete MSE results on 25 PDEs over ten independent runs. Lower is better.

PDE Problem	RoPINN	PINNsFormer	PINNsAgent	PINNForge	Rank	PINNacle Best
burgers_1d	7.31E-04 $\pm$ 6.13E-05	1.08E-03 $\pm$ 8.21E-06	6.52E-05 $\pm$ 1.55E-05	6.50E-05 $\pm$ 4.42E-07	1	7.90E-05
burgers_2d	2.40E-01 $\pm$ 3.12E-02	2.65E-01 $\pm$ 6.83E-02	1.98E-01 $\pm$ 1.66E-02	1.95E-01 $\pm$ 1.31E-02	1	1.09E-01
gray_scott_2d	8.64E-03 $\pm$ 4.53E-05	1.57E-02 $\pm$ 1.32E-03	4.36E-03 $\pm$ 3.08E-05	3.60E-03 $\pm$ 2.65E-04	1	4.32E-03
heat_2d_complex_geometry	4.00E-04 $\pm$ 9.10E-05	2.78E-03 $\pm$ 5.77E-05	1.84E-03 $\pm$ 9.95E-04	4.45E-04 $\pm$ 9.72E-05	2	8.53E-04
heat_2d_long_time	2.56E+00 $\pm$ 4.53E-01	3.83E+00 $\pm$ 1.64E-01	4.32E+00 $\pm$ 2.78E+00	1.13E+00 $\pm$ 4.21E-02	1	1.13E+00
heat_2d_multiscale	2.75E-05 $\pm$ 7.52E-06	5.57E-05 $\pm$ 9.93E-06	3.51E-05 $\pm$ 2.20E-05	1.19E-05 $\pm$ 2.16E-06	1	5.27E-05
heat_2d_varying_coefficient	6.24E-01 $\pm$ 4.53E-03	1.19E-02 $\pm$ 3.42E-03	5.34E-03 $\pm$ 3.85E-03	1.59E-03 $\pm$ 4.77E-04	1	1.76E-03
heat_5d	5.39E-07 $\pm$ 1.31E-07	1.87E-07 $\pm$ 1.91E-08	3.34E-07 $\pm$ 7.65E-08	3.21E-08 $\pm$ 3.26E-08	1	8.52E+00
heat_inverse_2d	4.31E-02 $\pm$ 7.31E-03	8.89E-02 $\pm$ 4.22E-02	6.32E-02 $\pm$ 8.63E-02	7.15E-03 $\pm$ 4.91E-03	1	5.66E-03
kuramoto_sivashinsky_1d	6.54E+00 $\pm$ 5.14E-01	5.13E+00 $\pm$ 4.51E-03	1.09E+00 $\pm$ 3.18E-02	1.03E+00 $\pm$ 8.40E-02	1	1.04E+00
navier_stokes_2d_backstep	9.64E-03 $\pm$ 5.13E-03	6.02E-04 $\pm$ 4.33E-05	4.96E-03 $\pm$ 1.45E-03	2.63E-04 $\pm$ 4.71E-05	1	3.37E-04
navier_stokes_2d_classic	2.88E-05 $\pm$ 3.61E-06	1.49E-05 $\pm$ 1.26E-05	8.48E-06 $\pm$ 6.90E-06	2.48E-06 $\pm$ 2.31E-06	1	2.33E-05
navier_stokes_2d_long_time	5.43E+02 $\pm$ 5.63E+00	7.43E+02 $\pm$ 1.31E+02	5.75E+02 $\pm$ 7.63E+00	5.04E+02 $\pm$ 4.95E+00	1	5.05E+02
poisson_2d_classic	8.57E-02 $\pm$ 6.74E-02	2.17E-01 $\pm$ 7.65E-02	6.34E-01 $\pm$ 1.43E-01	1.61E-02 $\pm$ 3.95E-04	1	5.00E-05
poisson_boltzmann_2d	7.91E-01 $\pm$ 7.14E-01	1.34E-02 $\pm$ 4.31E-03	6.97E-02 $\pm$ 7.43E-02	5.38E-03 $\pm$ 9.13E-04	1	6.99E-05
poisson_2d_many_area	2.89E+00 $\pm$ 3.51E-01	5.36E+00 $\pm$ 3.79E+00	3.00E+00 $\pm$ 1.01E+00	2.21E+00 $\pm$ 3.56E-01	1	1.83E+00
poisson_3d_complex_geometry	2.16E-01 $\pm$ 1.53E-02	8.16E-03 $\pm$ 5.58E-04	1.60E-02 $\pm$ 1.08E-02	1.04E-03 $\pm$ 2.34E-04	1	9.51E-04
poisson_5d	4.09E-07 $\pm$ 7.43E-08	2.92E-07 $\pm$ 1.99E-07	2.01E-06 $\pm$ 1.09E-06	5.64E-08 $\pm$ 3.95E-08	1	2.09E-06
poisson_inverse_2d	5.32E-03 $\pm$ 3.46E-03	1.51E-03 $\pm$ 8.65E-04	2.41E-02 $\pm$ 4.73E-02	2.57E-04 $\pm$ 1.29E-04	1	1.23E-04
wave_1d	4.19E-02 $\pm$ 4.38E-03	3.61E-02 $\pm$ 6.65E-03	3.25E-02 $\pm$ 3.45E-02	3.18E-02 $\pm$ 5.73E-03	1	3.01E-03
wave_2d_heterogeneous	4.63E-02 $\pm$ 6.73E-03	8.37E-02 $\pm$ 1.18E-02	5.17E-02 $\pm$ 8.03E-03	2.84E-02 $\pm$ 7.69E-04	1	2.99E-02
allen_cahn_1d	7.45E-08 $\pm$ 4.61E-08	6.02E-09 $\pm$ 1.08E-09	1.44E-08 $\pm$ 3.50E-09	1.13E-09 $\pm$ 2.26E-10	1	–
darcy_flow_2d	3.48E-09 $\pm$ 1.63E-09	1.35E-10 $\pm$ 2.90E-11	5.33E-07 $\pm$ 2.65E-07	1.64E-11 $\pm$ 3.28E-12	1	–
shallow_water_2d	5.78E-07 $\pm$ 7.13E-07	2.24E-07 $\pm$ 8.44E-08	6.81E-07 $\pm$ 1.25E-07	7.15E-08 $\pm$ 2.86E-08	1	–
kovasznay_flow_2d	6.13E-07 $\pm$ 4.31E-07	3.80E-08 $\pm$ 9.63E-09	3.82E-07 $\pm$ 3.66E-07	1.77E-09 $\pm$ 4.43E-10	1	–
Friedman Avg. Rank	3.04	2.96	2.96	1.04	–	–
Wilcoxon vs. PINNForge	+	+	+	–	–	–

Here, +, =, and – denote significantly better, statistically indistinguishable, and significantly worse performance of PINNForge, respectively, under paired Wilcoxon signed-rank tests across the 25 PDEs on $\log_{10}$ -transformed per-task mean MSE with Holm correction at $\alpha = 0.05$.

The Friedman test is applied to per-task ranks computed from the mean MSE over ten independent runs for each of the 25 PDEs and gives

$$\chi_F^2 = 42.696, \quad p = 2.86 \times 10^{-9},$$

indicating significant cross-task ranking differences. PINNForge has average rank 1.04, compared with 3.04, 2.96, and 2.96 for RoPINN, PINNsFormer, and PINNsAgent. Pairwise Wilcoxon tests across the 25 tasks on $\log_{10}$ -transformed per-task mean MSE, with Holm correction, yield $p_{\text{adj}} = 1.79 \times 10^{-7}$ for all three comparisons.

Median coefficients of variation are 0.177, 0.184, 0.497, and 0.182 for RoPINN, PINNsFormer, PINNsAgent, and PINNForge, respectively; PINNForge’s lower MSE is therefore not accompanied by an obvious loss of run-to-run stability.

B.7 COMPLETE COMPONENT ABLATION

The information channels available to each ablated variant are defined in Appendix A.6. To compare variants across PDEs whose MSE values span several orders of magnitude, we compute the task-wise error ratio

$$r(\mathcal{P}) = \frac{\text{MSE}_{\text{ablation}}(\mathcal{P})}{\text{MSE}_{\text{PINNForge}}(\mathcal{P})},$$

and report the median ratio across the 12 representative PDEs as an aggregate measure.

Table 14: Complete component-ablation results over 12 representative PDEs. Results are mean $\pm$ standard deviation. Lower MSE is better.

PDE	Characteristic	w/o Knowledge Guidance	w/o Execution Feedback to LLM	w/o Evolutionary Search	PINNForge
burgers_1d	Nonlinear / shock	7.42E-05 $\pm$ 6.31E-05	6.32E-04 $\pm$ 7.82E-05	2.78E-04 $\pm$ 3.24E-05	6.50E-05 $\pm$ 4.42E-07
burgers_2d	2D nonlinear transport	2.44E-01 $\pm$ 2.86E-02	6.71E-01 $\pm$ 8.37E-02	1.67E+00 $\pm$ 2.31E-01	1.95E-01 $\pm$ 1.31E-02
gray_scott_2d	Coupled reaction–diffusion	5.29E-03 $\pm$ 4.91E-04	5.36E-03 $\pm$ 5.62E-03	6.23E-03 $\pm$ 6.87E-04	3.60E-03 $\pm$ 2.65E-04
heat_2d_multiscale	Multiscale dynamics	3.81E-05 $\pm$ 4.28E-06	2.08E-04 $\pm$ 2.74E-04	1.30E-03 $\pm$ 1.66E-03	1.19E-05 $\pm$ 2.16E-06
heat_2d_varying_coefficient	Variable coefficient	2.18E-03 $\pm$ 2.03E-03	2.57E-03 $\pm$ 2.71E-04	4.15E-03 $\pm$ 4.86E-04	1.59E-03 $\pm$ 4.77E-04
heat_2d_complex_geometry	Complex geometry	1.90E-03 $\pm$ 2.41E-03	1.59E+00 $\pm$ 2.14E-01	1.97E-02 $\pm$ 2.77E-03	4.45E-04 $\pm$ 9.72E-05
poisson_3d_complex_geometry	3D complex geometry	8.79E-03 $\pm$ 9.38E-04	2.00E-02 $\pm$ 2.66E-03	2.49E-02 $\pm$ 3.18E-03	1.04E-03 $\pm$ 2.34E-04
poisson_5d	High-dimensional PDE	2.55E-07 $\pm$ 2.87E-07	1.24E-07 $\pm$ 1.41E-08	2.06E-05 $\pm$ 2.72E-06	5.64E-08 $\pm$ 3.95E-08
poisson_inverse_2d	Inverse problem	6.94E-04 $\pm$ 7.64E-05	3.72E-03 $\pm$ 4.58E-04	3.14E-04 $\pm$ 3.63E-04	2.57E-04 $\pm$ 1.29E-04
navier_stokes_2d_backstep	Nonlinear fluid dynamics	2.05E-03 $\pm$ 2.68E-04	4.50E-03 $\pm$ 6.13E-03	3.06E-03 $\pm$ 3.97E-03	2.63E-04 $\pm$ 4.71E-05
darcy_flow_2d	Heterogeneous	2.76E-10 $\pm$ 3.17E-10	2.97E-09 $\pm$ 4.06E-10	1.07E-08 $\pm$ 1.48E-09	1.64E-11 $\pm$ 3.28E-12
shallow_water_2d	Hyperbolic system	3.18E-07 $\pm$ 3.52E-08	2.32E-07 $\pm$ 2.69E-08	5.59E-07 $\pm$ 7.14E-08	7.15E-08 $\pm$ 2.86E-08
Median error ratio	–	3.74 $\times$	12.10 $\times$	10.10 $\times$	1.00 $\times$

Removing Knowledge Guidance raises the median error to 3.74 $\times$ that of full PINNForge. Because both $K_{\mathcal{P}}$ and M_g are removed, this reflects the knowledge-guidance channel as a whole rather than static prior knowledge alone. Removing Execution Feedback to LLM gives the largest degradation, 12.10 $\times$. Because this variant withholds candidate-specific execution evidence and therefore cannot form the execution-derived posterior memory, the remaining PDE-related prior knowledge and parent structures are insufficient to recover full-method performance under this setup. Removing Evolutionary Search raises the ratio to 10.10 $\times$ while retaining the candidate-evaluation budget, knowledge, and feedback. This is consistent with a benefit from parent-driven inheritance, but does not isolate all interactions among the retained channels.

B.8 TRAINING-BUDGET ACCOUNTING

We compare optimizer steps required to obtain the selected result on one PDE. Each baseline run trains five independently initialized candidates for 20,000 steps each and selects the lowest-MSE

candidate, giving

$$5 \times 20,000 = 100,000.$$

PINNForge evaluates eight Generation-0 candidates and at most nine later generations with five candidates each, for a maximum of

$$8 + 9 \times 5 = 53.$$

At 1,000 steps per candidate, low-fidelity search uses at most 53,000 steps. Independently retraining the Global Top-3 for 10,000 steps each adds 30,000, so the maximum total is

$$53,000 + 30,000 = 83,000.$$

Table 15: Optimizer-step budget for obtaining the selected result on one PDE.

Method	Selection Protocol	Budget
RoPINN	Best of 5 independently initialized 20K candidates	100K
PINNsFormer	Best of 5 independently initialized 20K candidates	100K
PINNsAgent	Best of 5 independently initialized 20K candidates	100K
PINNForge	$53 \times 1K$ LF + $Top-3 \times 10K$ HF	$\leq 83K$

This accounting concerns PINN optimizer steps rather than total computational cost. The cost of an individual step can differ across architectures, sampling configurations, and optimizers. LLM inference overhead, wall-clock time, and monetary cost are not included.

C COMPLETE SEARCH TRAJECTORIES

Table C.1 reports single-run search trajectories for all 25 PDEs using the fixed reference evaluation points in Appendix B.4. Generation-0 Best is the lowest initial MSE, Best LF the lowest MSE found during low-fidelity search, and Final HF the lowest MSE among independently trained Global Top-3 candidates. The improvement ratios are computed as

$$G0 \rightarrow \text{Best LF} = \frac{MSE_{G0}}{MSE_{\text{Best LF}}},$$

$$G0 \rightarrow \text{Final HF} = \frac{MSE_{G0}}{MSE_{\text{Final HF}}}.$$

Table C.1: Complete single-run search trajectories across 25 PDE problems.

PDE	Generation-0 Best MSE	Best LF MSE	Final HF MSE	$G0 \rightarrow \text{Best LF}$	$G0 \rightarrow \text{Final HF}$
burgers_1d	3.64E-03	2.05E-04 (G8)	6.51E-05	17.74×	55.82×
burgers_2d	2.49E-01	2.13E-01 (G5)	1.78E-01	1.17×	1.39×
gray_scott_2d	4.37E-03	4.03E-03 (G9)	4.07E-03	1.08×	1.07×
heat_2d_complex_geometry	1.28E-02	3.26E-03 (G8)	4.63E-04	3.92×	27.64×
heat_2d_long_time	1.13E+00	1.10E+00 (G5)	1.08E+00	1.03×	1.05×
heat_2d_multiscale	3.01E-02	9.140E-05 (G9)	1.014E-05	330.08×	2,975.09×
heat_2d_varying_coefficient	1.12E-02	2.85E-03 (G9)	2.18E-03	3.93×	5.13×
heat_5d	1.31E-05	3.85E-07 (G9)	8.99E-09	33.90×	1,452.78×
heat_inverse_2d	1.36E-02	5.73E-03 (G6)	3.68E-03	2.36×	3.68×

PDE	Generation-0 Best MSE	Best LF MSE	Final HF MSE	G0 → Best LF	G0 → Final HF
kuramoto_sivashinsky_1d	1.34E+00	1.05E+00 (G7)	1.16E+00	1.28×	1.16×
navier_stokes_2d_backstep	2.64E-03	7.68E-04 (G9)	2.30E-04	3.44×	11.48×
navier_stokes_2d_classic	1.86E-03	1.61E-05 (G9)	2.35E-06	115.60×	792.09×
navier_stokes_2d_long_time	5.09E+02	5.08E+02 (G7)	4.97E+02	1.00×	1.02×
poisson_2d_classic	9.77E-02	8.94E-02 (G8)	1.64E-02	1.09×	5.95×
poisson_boltzmann_2d	1.04E-01	1.23E-02 (G9)	6.03E-03	8.45×	17.16×
poisson_2d_many_area	3.27E+00	2.79E+00 (G7)	1.97E+00	1.17×	1.66×
poisson_3d_complex_geometry	8.96E-03	4.48E-03 (G8)	8.83E-04	2.00×	10.14×
poisson_5d	6.44E-06	1.01E-07 (G9)	5.65E-08	63.58×	114.03×
poisson_inverse_2d	6.22E-03	4.32E-04 (G8)	1.45E-04	14.39×	42.80×
wave_1d	7.43E-02	5.57E-02 (G6)	3.15E-02	1.34×	2.36×
wave_2d_heterogeneous	7.85E-02	3.85E-02 (G7)	2.96E-02	2.03×	2.65×
allen_cahn_1d	6.37E-08	1.08E-08 (G9)	1.20E-09	5.89×	53.1×
darcy_flow_2d	6.94E-09	3.14E-10 (G8)	1.53E-11	22.11×	453.59×
shallow_water_2d	2.73E-09	2.33E-10 (G9)	9.83E-11	11.68×	27.71×
kovasznyai_flow_2d	2.00E-05	4.95E-08 (G9)	2.59E-09	404.28×	7,722.18×

D BENCHMARK PDE DEFINITIONS

This appendix summarizes the governing equations, computational domains, physical parameters, initial and boundary conditions, and reference solutions of the 25 PDE problems used in our experiments. The formulations follow the corresponding benchmark settings used for training and evaluation.

We use

$$\Delta u = \sum_{i=1}^d \frac{\partial^2 u}{\partial x_i^2}, \quad \partial_n u = \nabla u \cdot \mathbf{n},$$

where $\mathbf{n}$ denotes the outward unit normal. Unless otherwise specified, all parameters take the values given below.

D.1 ONE-DIMENSIONAL BURGERS EQUATION (`BURGERS_1D`)

The governing equation is

$$u_t + uu_x - \nu u_{xx} = 0, \quad (x, t) \in [-1, 1] \times [0, 1],$$

with viscosity

$$\nu = \frac{0.01}{\pi}.$$

The initial condition is

$$u(x, 0) = -\sin(\pi x),$$

and homogeneous Dirichlet boundary conditions are imposed at both spatial boundaries:

$$u(-1, t) = u(1, t) = 0.$$

D.2 TWO-DIMENSIONAL BURGERS SYSTEM (`BURGERS_2D`)

Let

$$\mathbf{u} = (u, v)^\top.$$

The coupled two-dimensional Burgers system is

$$\begin{aligned} u_t + uu_x + vv_y - \nu(u_{xx} + u_{yy}) &= 0, \\ v_t + uv_x + vv_y - \nu(v_{xx} + v_{yy}) &= 0, \end{aligned}$$

on

$$(x, y, t) \in [0, 4]^2 \times [0, 1],$$

with

$$\nu = 10^{-3}.$$

The initial fields are prescribed by the benchmark data:

$$u(x, y, 0) = u_0(x, y), \quad v(x, y, 0) = v_0(x, y),$$

where u_0 and v_0 are interpolated from `burgers2d_init_u_0.dat` and `burgers2d_init_v_0.dat`, respectively.

Both components are periodic in the two spatial directions:

$$\begin{aligned} u(0, y, t) &= u(4, y, t), & v(0, y, t) &= v(4, y, t), \\ u(x, 0, t) &= u(x, 4, t), & v(x, 0, t) &= v(x, 4, t). \end{aligned}$$

D.3 GRAY-SCOTT REACTION-DIFFUSION SYSTEM (`GRAY_SCOTT_2D`)

The governing equations are

$$\begin{aligned} u_t &= \varepsilon_1 \Delta u + b(1 - u) - uv^2, \\ v_t &= \varepsilon_2 \Delta v - dv + uv^2, \end{aligned}$$

on

$$(x, y, t) \in [-1, 1]^2 \times [0, 200],$$

with

$$b = 0.04, \quad d = 0.1, \quad \varepsilon_1 = 10^{-5}, \quad \varepsilon_2 = 5 \times 10^{-6}.$$

The initial conditions are

$$u(x, y, 0) = 1 - \exp \left[-80 \left((x + 0.05)^2 + (y + 0.02)^2 \right) \right],$$

and

$$v(x, y, 0) = \exp \left[-80 \left((x - 0.05)^2 + (y - 0.02)^2 \right) \right].$$

Following the benchmark protocol, no additional spatial boundary loss is imposed.

D.4 HEAT EQUATION ON A PERFORATED DOMAIN (`HEAT_2D_COMPLEX_GEOMETRY`)

The governing equation is

$$u_t - \Delta u = 0, \quad (x, y, t) \in \Omega \times [0, 3].$$

The spatial domain is the rectangle

$$R = [-8, 8] \times [-12, 12],$$

with 17 circular holes removed:

$$\Omega = R \setminus \left(\bigcup_{c \in \mathcal{C}_1} B_1(c) \cup \bigcup_{c \in \mathcal{C}_{0.4}} B_{0.4}(c) \right),$$

where

$$\begin{aligned} \mathcal{C}_1 = \{ & (-4, -3), (4, -3), (-4, 3), (4, 3), \\ & (-4, -9), (4, -9), (-4, 9), (4, 9), \\ & (0, 0), (0, 6), (0, -6) \}, \end{aligned}$$

and

$$\mathcal{C}_{0.4} = \{(-3.2, -6), (-3.2, 6), (3.2, -6), (3.2, 6), (-3.2, 0), (3.2, 0)\}.$$

The initial condition is

$$u(x, y, 0) = 0.$$

Robin boundary conditions are imposed as

$$\partial_n u = \begin{cases} 5 - u, & \text{on the boundaries of the radius-1 holes,} \\ 1 - u, & \text{on the boundaries of the radius-0.4 holes,} \\ 0.1 - u, & \text{on the outer rectangular boundary.} \end{cases}$$

D.5 LONG-TIME NONLINEAR HEAT EQUATION (`HEAT_2D_LONG_TIME`)

The governing equation is

$$u_t - 10^{-3}(u_{xx} + u_{yy}) - 5 \sin(u^2) \left(1 + 2 \sin \frac{\pi t}{4} \right) \sin(4\pi x) \sin(2\pi y) = 0,$$

for

$$(x, y, t) \in [0, 1]^2 \times [0, 100].$$

The initial condition is

$$u(x, y, 0) = \sin(4\pi x) \sin(3\pi y).$$

Homogeneous Dirichlet boundary conditions are imposed on the spatial boundary:

$$u(x, y, t) = 0, \quad (x, y) \in \partial[0, 1]^2.$$

D.6 MULTISCALE ANISOTROPIC HEAT EQUATION (`HEAT_2D_MULTISCALE`)

The governing equation is

$$u_t - \frac{1}{(500\pi)^2} u_{xx} - \frac{1}{\pi^2} u_{yy} = 0,$$

on

$$(x, y, t) \in [0, 1]^2 \times [0, 5].$$

The initial condition is

$$u(x, y, 0) = \sin(20\pi x) \sin(\pi y).$$

Homogeneous Dirichlet boundary conditions are imposed:

$$u(x, y, t) = 0, \quad (x, y) \in \partial[0, 1]^2.$$

The analytical reference solution is

$$u^*(x, y, t) = \sin(20\pi x) \sin(\pi y) \exp \left[- \left(\frac{(20\pi)^2}{(500\pi)^2} + \frac{\pi^2}{\pi^2} \right) t \right].$$

D.7 VARYING-COEFFICIENT HEAT EQUATION (`HEAT_2D_VARYING_COEFFICIENT`)

The governing equation used in the benchmark is

$$u_t - \kappa(x, y) \Delta u = f(x, y, t),$$

on

$$(x, y, t) \in [0, 1]^2 \times [0, 5],$$

where $\kappa(x, y)$ is the heterogeneous coefficient field stored in `heat_2d_coef_256.dat` and evaluated by nearest-neighbour interpolation.

The source term is

$$f(x, y, t) = 200 \sin(\pi x) \sin(5\pi y) \sin(\pi t).$$

The initial condition is

$$u(x, y, 0) = 0,$$

and homogeneous Dirichlet boundary conditions are imposed:

$$u|_{\partial[0,1]^2} = 0.$$

D.8 FIVE-DIMENSIONAL HEAT EQUATION (HEAT_5D)

Let

$$\Omega = \{\mathbf{x} \in \mathbb{R}^5 : \|\mathbf{x}\|_2 \leq 1\}.$$

The governing equation is

$$u_t - \frac{1}{5}\Delta u = f(\mathbf{x}, t), \quad (\mathbf{x}, t) \in \Omega \times [0, 1],$$

where

$$f(\mathbf{x}, t) = -\frac{\|\mathbf{x}\|_2^2}{5} \exp\left(\frac{\|\mathbf{x}\|_2^2}{2} + t\right).$$

Define

$$g(\mathbf{x}, t) = \exp\left(\frac{\|\mathbf{x}\|_2^2}{2} + t\right).$$

The initial condition is

$$u(\mathbf{x}, 0) = g(\mathbf{x}, 0),$$

and the Neumann boundary condition is

$$\partial_n u = g(\mathbf{x}, t), \quad \mathbf{x} \in \partial\Omega.$$

The reference solution is

$$u^*(\mathbf{x}, t) = g(\mathbf{x}, t).$$

D.9 INVERSE HEAT EQUATION (HEAT_INVERSE_2D)

The unknowns are the state $u(x, y, t)$ and the spatially varying diffusion coefficient $a(x, y)$. They satisfy

$$u_t - \nabla \cdot (a \nabla u) = f(x, y, t),$$

on

$$(x, y, t) \in [-1, 1]^2 \times [0, 1].$$

The manufactured reference fields are

$$u^*(x, y, t) = e^{-t} \sin(\pi x) \sin(\pi y),$$

and

$$a^*(x, y) = 2 + \sin(\pi x) \sin(\pi y).$$

The corresponding source term is

$$f(x, y, t) = e^{-t} \left[(4\pi^2 - 1) \sin(\pi x) \sin(\pi y) + \pi^2 (2 \sin^2(\pi x) \sin^2(\pi y) - \cos^2(\pi x) \sin^2(\pi y) - \sin^2(\pi x) \cos^2(\pi y)) \right].$$

Since

$$a^* = 2$$

on the spatial boundary, the coefficient boundary condition is

$$a(x, y) = 2, \quad (x, y) \in \partial[-1, 1]^2.$$

The inverse problem additionally uses 2500 noisy state observations:

$$u_{\text{obs}} = u^* + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.1^2).$$

D.10 KURAMOTO–SIVASHINSKY EQUATION (KURAMOTO_SIVASHINSKY_1D)

The governing equation is

$$u_t + \alpha u u_x + \beta u_{xx} + \gamma u_{xxx} = 0,$$

on

$$(x, t) \in [0, 2\pi] \times [0, 1],$$

where

$$\alpha = \frac{100}{16}, \quad \beta = \frac{100}{16^2}, \quad \gamma = \frac{100}{16^4}.$$

The initial condition is

$$u(x, 0) = \cos(x)(1 + \sin x).$$

Following the benchmark protocol, the spatial domain is treated as periodic, while only the initial condition is explicitly included in the constraint loss.

D.11 NAVIER–STOKES FLOW OVER A BACKWARD-FACING STEP (NAVIER_STOKES_2D_BACKSTEP)

Let

$$\mathbf{u} = (u, v)^\top.$$

The steady incompressible Navier–Stokes equations are

$$(\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \nu \Delta \mathbf{u} = \mathbf{0},$$

together with

$$\nabla \cdot \mathbf{u} = 0,$$

where

$$\nu = 0.01.$$

The computational domain is

$$\Omega = ([0, 4] \times [0, 2]) \setminus ([0, 2] \times [1, 2]).$$

At the inlet $x = 0$, $0 \leq y \leq 1$,

$$u(0, y) = 4y(1 - y), \quad v(0, y) = 0.$$

At the outlet $x = 4$,

$$p(4, y) = 0.$$

The no-slip condition is imposed on the remaining solid walls:

$$u = v = 0.$$

D.12 LID-DRIVEN CAVITY FLOW (`NAVIER_STOKES_2D_CLASSIC`)

This problem corresponds to the benchmark case `navier_stokes_2d_C`.

The steady incompressible Navier–Stokes equations are

$$(\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \nu \Delta \mathbf{u} = \mathbf{0},$$

and

$$\nabla \cdot \mathbf{u} = 0,$$

on

$$(x, y) \in [0, 1]^2.$$

The viscosity is

$$\nu = 0.01,$$

corresponding to

$$\text{Re} = 100.$$

On the moving top wall,

$$u(x, 1) = 4x(1 - x), \quad v(x, 1) = 0.$$

On the other three walls,

$$u = v = 0.$$

Pressure is fixed by the gauge condition

$$p(0, 0) = 0.$$

D.13 LONG-TIME INCOMPRESSIBLE NAVIER-STOKES EQUATION (NAVIER-STOKES_2D_LONG_TIME)

The governing equations are

$$\mathbf{u}_t + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \nu \Delta \mathbf{u} = \mathbf{f}(x, y, t),$$

and

$$\nabla \cdot \mathbf{u} = 0,$$

on

$$(x, y, t) \in [0, 2] \times [0, 1] \times [0, 5],$$

with

$$\nu = 0.01.$$

The forcing term is

$$\mathbf{f}(x, y, t) = \begin{pmatrix} 0 \\ -\sin(\pi x) \sin(\pi y) \sin(\pi t) \end{pmatrix}.$$

At the inlet $x = 0$,

$$u(0, y, t) = \sin(\pi y) [\sin(\pi t) + \sin(3\pi t) + \sin(5\pi t)],$$

and

$$v(0, y, t) = 0.$$

At the outlet $x = 2$,

$$p(2, y, t) = 0.$$

On the remaining spatial boundaries,

$$u = v = 0.$$

The initial condition is

$$u(x, y, 0) = v(x, y, 0) = p(x, y, 0) = 0.$$

D.14 CLASSICAL POISSON EQUATION ON A PERFORATED DOMAIN (POISSON_2D_CLASSIC)

The governing equation is

$$\Delta u = 0 \quad \text{in } \Omega,$$

where

$$\Omega = \left[-\frac{1}{2}, \frac{1}{2}\right]^2 \setminus \bigcup_{c \in \mathcal{C}} B_{0.1}(c),$$

and

$$\mathcal{C} = \{(0.3, 0.3), (-0.3, 0.3), (0.3, -0.3), (-0.3, -0.3)\}.$$

The Dirichlet boundary conditions are

$$u = 1 \quad \text{on the outer rectangular boundary,}$$

and

$$u = 0 \quad \text{on the four circular boundaries.}$$

D.15 SCREENED POISSON–BOLTZMANN PROBLEM (POISSON_BOLTZMANN_2D)

The benchmark considers the screened Poisson equation

$$-\Delta u + k^2 u = f(x, y),$$

with

$$k = 8.$$

The source term is

$$f(x, y) = A (\mu_1^2 + x^2 + \mu_2^2 + y^2) \sin(\mu_1 \pi x) \sin(\mu_2 \pi y),$$

where

$$A = 10, \quad \mu_1 = 1, \quad \mu_2 = 4.$$

The spatial domain is

$$\Omega = [-1, 1]^2 \setminus \bigcup_{j=1}^4 B_{r_j}(c_j),$$

where

$$\begin{aligned} (c_1, r_1) &= ((0.5, 0.5), 0.2), \\ (c_2, r_2) &= ((0.4, -0.4), 0.4), \\ (c_3, r_3) &= ((-0.2, -0.7), 0.1), \\ (c_4, r_4) &= ((-0.6, 0.5), 0.3). \end{aligned}$$

The Dirichlet conditions are

$$u = 0.2 \quad \text{on the outer boundary,}$$

and

$$u = 1 \quad \text{on the circular boundaries.}$$

D.16 MULTISCALE MULTI-AREA POISSON EQUATION (`POISSON_2D_MANY_AREA`)

The governing equation is

$$a(x, y)\Delta u + f(x, y) = 0, \quad (x, y) \in [-10, 10]^2.$$

The domain is partitioned into 5×5 rectangular subdomains. Within subdomain Ω_{ij} ,

$$a(x, y) = a_{ij},$$

where a_{ij} is read from `poisson_a_coef.dat`.

The local forcing is defined by the tabulated coefficients in `poisson_f_coef.dat`:

$$f(x, y) = c_{ij,00} + \sum_{r=0}^1 \sum_{s=0}^1 c_{ij,rs} \sin\left(\frac{\pi r \xi}{h_x}\right) \sin\left(\frac{\pi s \eta}{h_y}\right), \quad (x, y) \in \Omega_{ij},$$

where (ξ, η) denote the local coordinates within Ω_{ij} .

The Robin boundary condition is

$$\partial_n u = -u \quad \text{on } \partial[-10, 10]^2.$$

D.17 THREE-DIMENSIONAL POISSON EQUATION ON A COMPLEX GEOMETRY (`POISSON_3D_COMPLEX_GEOMETRY`)

The governing equation is

$$-\mu(z)\Delta u + \kappa(z)u = f(x, y, z),$$

in a unit cube with four spherical holes:

$$\Omega = [0, 1]^3 \setminus \bigcup_{j=1}^4 B_{r_j}(\mathbf{c}_j),$$

where

$$\begin{aligned} (\mathbf{c}_1, r_1) &= ((0.4, 0.3, 0.6), 0.2), \\ (\mathbf{c}_2, r_2) &= ((0.6, 0.7, 0.6), 0.2), \\ (\mathbf{c}_3, r_3) &= ((0.2, 0.8, 0.7), 0.1), \\ (\mathbf{c}_4, r_4) &= ((0.6, 0.2, 0.3), 0.1). \end{aligned}$$

The coefficients are

$$\mu(z) = 1,$$

and

$$\kappa(z) = \begin{cases} 8^2, & z < 0.5, \\ 10^2, & z \geq 0.5. \end{cases}$$

Let

$$r^2 = x^2 + y^2 + z^2.$$

The forcing term is

$$f(x, y, z) = 20 \exp [\sin(\pi x) + \sin(10\pi y) + \sin(5\pi z)] \frac{r^2 - 1}{r^2 + 1} + 100 \sin(\pi x) \sin(10\pi y) \sin(5\pi z).$$

A homogeneous Neumann condition is imposed on the complete boundary:

$$\partial_n u = 0 \quad \text{on } \partial\Omega.$$

D.18 FIVE-DIMENSIONAL POISSON EQUATION (POISSON_5D)

The governing equation is

$$-\Delta u = \frac{\pi^2}{4} \sum_{i=1}^5 \sin\left(\frac{\pi x_i}{2}\right), \quad \mathbf{x} \in [0, 1]^5.$$

The analytical reference solution is

$$u^*(\mathbf{x}) = \sum_{i=1}^5 \sin\left(\frac{\pi x_i}{2}\right).$$

The Dirichlet boundary condition is

$$u = u^* \quad \text{on } \partial[0, 1]^5.$$

D.19 INVERSE POISSON EQUATION (POISSON_INVERSE_2D)

The unknowns are $u(x, y)$ and the spatially varying diffusion coefficient $a(x, y)$, satisfying

$$-\nabla \cdot (a \nabla u) = f(x, y), \quad (x, y) \in [0, 1]^2.$$

The manufactured reference fields are

$$u^*(x, y) = \sin(\pi x) \sin(\pi y),$$

and

$$a^*(x, y) = \frac{1}{1 + x^2 + y^2 + (x - 1)^2 + (y - 1)^2}.$$

Writing

$$a = a^*(x, y),$$

the corresponding source term is

$$\begin{aligned} f(x, y) = & 2\pi^2 a \sin(\pi x) \sin(\pi y) \\ & + 2\pi a^2 \left[(2x - 1) \cos(\pi x) \sin(\pi y) \right. \\ & \left. + (2y - 1) \sin(\pi x) \cos(\pi y) \right]. \end{aligned}$$

The diffusion coefficient is prescribed on the spatial boundary:

$$a = a^* \quad \text{on } \partial[0, 1]^2.$$

The inverse problem uses a 50×50 grid of noisy state observations:

$$u_{\text{obs}} = u^* + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.1^2).$$

D.20 ONE-DIMENSIONAL WAVE EQUATION (`WAVE_1D`)

The governing equation is

$$u_{tt} - 4u_{xx} = 0, \quad (x, t) \in [0, 1] \times [0, 1].$$

The initial conditions are

$$u(x, 0) = \sin(\pi x) + \frac{1}{2} \sin(4\pi x),$$

and

$$u_t(x, 0) = 0.$$

The spatial boundary condition is

$$u(0, t) = u(1, t) = 0.$$

The analytical reference solution is

$$u^*(x, t) = \sin(\pi x) \cos(2\pi t) + \frac{1}{2} \sin(4\pi x) \cos(8\pi t).$$

D.21 HETEROGENEOUS TWO-DIMENSIONAL WAVE EQUATION (`WAVE_2D_HETEROGENEOUS`)

The governing equation is

$$\Delta u - \frac{1}{c(x, y)} u_{tt} = 0,$$

or equivalently,

$$u_{tt} = c(x, y) \Delta u,$$

on

$$(x, y, t) \in [-1, 1]^2 \times [0, 5].$$

The heterogeneous coefficient $c(x, y)$ is defined by interpolation of the Gaussian-random-field realization stored in `darcy_2d_coef_256.dat`.

The initial conditions are

$$u(x, y, 0) = \exp \left[-\frac{(x + 0.5)^2 + y^2}{2(0.3)^2} \right],$$

and

$$u_t(x, y, 0) = 0.$$

A homogeneous Neumann condition is imposed on the spatial boundary:

$$\partial_n u = 0 \quad \text{on } \partial[-1, 1]^2.$$

D.22 ONE-DIMENSIONAL ALLEN–CAHN EQUATION (`ALLEN_CAHN_1D`)

The governing equation is

$$u_t - \varepsilon^2 u_{xx} + u^3 - u = 0,$$

on

$$(x, t) \in [-1, 1] \times [0, 0.25],$$

with

$$\varepsilon = 0.1.$$

The initial condition is

$$u(x, 0) = \tanh \left(\frac{x}{\sqrt{2}\varepsilon} \right).$$

The Dirichlet boundary condition is obtained from the stationary reference solution:

$$u(\pm 1, t) = \tanh \left(\frac{\pm 1}{\sqrt{2}\varepsilon} \right).$$

The analytical reference solution is

$$u^*(x, t) = \tanh \left(\frac{x}{\sqrt{2}\varepsilon} \right).$$

D.23 VARIABLE-COEFFICIENT DARCY EQUATION (`DARCY_FLOW_2D`)

The governing equation is

$$-\nabla \cdot (a(x, y) \nabla u) = f(x, y), \quad (x, y) \in [0, 1]^2.$$

The heterogeneous coefficient is

$$a(x, y) = 2 + \frac{1}{2} \sin(2\pi x) \sin(3\pi y).$$

The manufactured reference solution is

$$u^*(x, y) = \sin(\pi x) \sin(\pi y).$$

The source term is defined consistently as

$$f(x, y) = -\nabla \cdot (a(x, y) \nabla u^*(x, y)).$$

Equivalently,

$$f = -(a_x u_x^* + a_y u_y^* + a \Delta u^*).$$

The homogeneous Dirichlet boundary condition is

$$u = 0 \quad \text{on } \partial[0, 1]^2.$$

D.24 TWO-DIMENSIONAL SHALLOW-WATER SYSTEM (`SHALLOW_WATER_2D`)

Let h denote the water depth, and let q_x and q_y denote the two momentum components.

The conservative shallow-water equations are

$$h_t + (q_x)_x + (q_y)_y = 0,$$

$$(q_x)_t + \left(\frac{q_x^2}{h} + \frac{gh^2}{2} \right)_x + \left(\frac{q_x q_y}{h} \right)_y = 0,$$

and

$$(q_y)_t + \left(\frac{q_x q_y}{h} \right)_x + \left(\frac{q_y^2}{h} + \frac{gh^2}{2} \right)_y = 0,$$

on

$$(x, y, t) \in [0, 1]^2 \times [0, 0.1],$$

with gravitational acceleration

$$g = 9.81.$$

The initial state is spatially uniform:

$$(h, q_x, q_y)(x, y, 0) = (1, 0.1, -0.05).$$

All three variables are periodic in both spatial directions:

$$U(0, y, t) = U(1, y, t),$$

and

$$U(x, 0, t) = U(x, 1, t),$$

where

$$U = (h, q_x, q_y)^\top.$$

Water-depth positivity is enforced through

$$h \geq h_{\min}, \quad h_{\min} = 10^{-3}.$$

The reference solution is the constant state

$$U^* = (1, 0.1, -0.05)^\top.$$

D.25 KOVASZNAY FLOW (`KOVASZNAY_FLOW_2D`)

The steady incompressible Navier–Stokes equations are

$$(\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \nu \Delta \mathbf{u} = \mathbf{0},$$

together with

$$\nabla \cdot \mathbf{u} = 0.$$

The computational domain is

$$(x, y) \in [-0.5, 1] \times [-0.5, 1.5].$$

The Reynolds number and viscosity are

$$\text{Re} = 40, \quad \nu = \frac{1}{40}.$$

Define

$$\lambda = \frac{\text{Re}}{2} - \sqrt{\frac{\text{Re}^2}{4} + 4\pi^2}.$$

The analytical reference solution is

$$u^*(x, y) = 1 - e^{\lambda x} \cos(2\pi y),$$

$$v^*(x, y) = \frac{\lambda}{2\pi} e^{\lambda x} \sin(2\pi y),$$

and

$$p^*(x, y) = \frac{1}{2} (1 - e^{2\lambda x}).$$

The analytical values are imposed on the rectangular boundary:

$$(u, v, p) = (u^*, v^*, p^*) \quad \text{on } \partial\Omega.$$

Pressure is additionally anchored by

$$p(0, 0) = 0.$$

D.26 NOTE ON THE EXCLUDED PINNACLE WAVE2D-MS PROBLEM

Among the PINNacle problems, we exclude the 2D multi-scale long-time wave equation (Wave2d-MS) because its published mathematical specification is internally inconsistent. PINNacle defines the governing equation as

$$u_{tt} - (u_{xx} + a^2 u_{yy}) = 0, \quad a = \sqrt{2},$$

while the reported reference solution is

$$\sinh(\pi x) \sinh(\pi y) \cos(\sqrt{3}\pi t).$$

Direct substitution gives

$$-6\pi^2 u^* \neq 0.$$

showing that the reported reference solution does not satisfy the stated governing equation.

We further observed that the reference solution in the released PINNacle implementation differs from the one reported in the paper and, under the released default parameters, is also inconsistent with the stated PDE. To avoid evaluating PINN methods against an ambiguous PDE–reference pair, we therefore exclude Wave2d-MS from our benchmark rather than modifying its original definition. All other PINNacle problems used in this work follow the corresponding benchmark formulations described above.

E BROADER IMPACT AND LIMITATIONS

PINNForge aims to reduce manual trial and error while making automated PINN design more traceable through structured configurations and execution evidence. Lower reference error, however, does not guarantee physical correctness or generalization. The same fixed reference points are reused for search control, final selection, and reporting, so the reported MSE measures this benchmark protocol rather than held-out generalization; adaptive reuse may favor configurations tuned to those points. Designed solvers may also fail in unsampled regions, out-of-distribution regimes, or unseen parameter settings. Independent evaluation points, physical-consistency checks, and domain-expert assessment would provide stronger validation.

The experiments use a single LLM backbone, Qwen3.8-27B, and no provider-side sampling seed, so they do not establish cross-backbone robustness or exact replay of individual search trajectories. PINNForge also incurs LLM inference overhead through multi-round generation and analysis. The optimizer-step comparison in Section 4.3 therefore measures PINN training cost rather than total computational or monetary cost. Future evaluation should consider benchmark accuracy together with LLM inference, wall-clock time, and overall resource use.